Chapter 17, Problem 57P

Problem

The coefficients of the trigonometric Fourier representation of a function are:

$$b_n = 0, \qquad a_n = \frac{6}{n^3 - 2}, \qquad n = 0, 1, 2, \cdots$$

If $\omega n = 50n$, find the exponential Fourier series for the function.

Step-by-step solution

Step 1 of 1

Given

$$b_n = 0,$$

$$a_n = \frac{6}{n^3 - 2}, \quad n = 0, 1, 2, \ldots..$$

$$\omega_n = n\omega_o = 50n$$

$$\omega_o = 50$$

We know that $C_n = \dfrac{a_n - jb_n}{2}$

$$C_n = \frac{\frac{6}{n^3 - 2} - j(0)}{2}$$

$$C_n = \frac{6}{2\left(n^3 - 2\right)}$$

$$C_n = \frac{3}{n^3 - 2}$$

$$a_0 = \frac{6}{0^3 - 2}$$

$$a_0 = -3$$

We know that $a_0 = C_0$

Therefore Fourier series expression is

$$\boxed{f(t) = -3 + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{3}{n^3 - 2} e^{-j50nt}}$$